Md Aquib Molla

Curriculum Vitae

mollaaquib@gmail.com
0000-0003-0416-1349
AquibMolla
Personal Website

Research Profile

I am a theoretical physicist working in nonequilibrium statistical physics. My research focuses on stochastic processes in spatially extended systems, including foraging dynamics, first-passage phenomena, quantum walks, percolation in excitable media, and active matter. Using analytical approaches and numerical simulations, I investigate how geometry, motion, memory and resource constraints shape transport, survival, and collective behaviour in complex systems.

Academic Metrics

5 peer-reviewed publications

3 corresponding-author publications

1 preprint on arXiv

3 ongoing research projects

International research visit to Germany

Multiple international talks and posters

Education

2021–2026 (expected) **PhD in Physics**, *Vidyasagar College, University of Calcutta*
Working title: *Effect of Environment and Constraints on Spreading in Complex Systems*

2018–2020 **MSc in Physics**, *University of Calcutta*
CGPA: 7.15

2015–2018 **BSc in Physics**, *University of Calcutta*

Research Interests

- Nonequilibrium Statistical Physics
- Active Matter
- Stochastic Processes
- First-Passage Phenomena
- Search and Foraging Theory
- Quantum Walks
- Collective Dynamics
- Percolation and Disordered Systems

Scientific Mobility

2025 **Research Visit**, *University of Potsdam, Germany*
Hosted by Prof. Ralf Metzler. Invited seminar: *From Cardiac Signals to Wandering Foragers: Physics of Living Systems*

2025 **Workshop**, *Driven and Active Amorphous Matter*, Göttingen, Germany

2025 **School**, *Leuven School on Nonequilibrium Statistical Mechanics*, KU Leuven, Belgium

2025 **Invited Seminar**, *Molecular biophysics at the transition state*, University of Trento, Italy (online)

2024 **Workshop**, *Discrete Integrable Systems*, ICTS-TIFR, *Bengaluru*, India

2024 **Conference**, *Data Dynamics Summit*, IISER, *Pune*, India

2023 **Seminer**, *One Day Student Seminar*, Behala College, Kolkata, India

2022 **Wrokshop**, *First-Passage Percolation and Related Models*, ICTS-TIFR, *Bengaluru*, India

Publications

Journal Articles

2026 **M. A. Molla**, S. Goswami, P. Sen, *Forager with intermittent rest: Better for survival?* Physical Review E **113**, 034139.

2026 **M. A. Molla**, S. Goswami, *Does Partial Consumption Help Foraging?* Accepted in Physica A.

2026 P. Maji, **M. A. Molla**, K. Dey et al., *Percolative Instabilities and Sparse-Limit Fractality in 1T-TaS_2* Physical Review B **113**, 115110.

2024 **M. A. Molla**, S. Goswami, *An Insight into Heart-like Systems with Percolation* Physics Letters A **518**, 129695.

2023 **M. A. Molla**, S. Goswami, *Quantum Walker in Presence of a Moving Detector* Physica A **620**, 128775.

Preprints

2026 **M. A. Molla**, S. Goswami, *Moving Detector Quantum Walk with Random Relocation* arXiv:2604.03593.

Under Preparaion

2026 **M. A. Molla** *Does Being Selfish Always Help?: A Simple Model of Cooperation and Competition,* Under preparation.

Ongoing Research Projects

Project 1 Active particles with memory effects and stochastic dynamics

Project 2 Cooperation, competition and strategic foraging in adaptive environments

Project 3 Kinetic and coarse-grained descriptions of search processes and collective dynamics

Selected Talks

2025 University of Potsdam, **Germany**, *From Cardiac Signals to Wandering Foragers: Physics of Living Systems*

2025 University of Trento, **Italy**, *Percolation in Excitable Media: Insights into Signal Propagation in Heart-like Systems*

2025 KU Leuven, **Belgium**, *Cardiac Dynamics through Percolation and Nonequilibrium Physics*

2025 Vidyasagar College, India, *Does Being Selfish Always Help?*

2024 ICTS-TIFR, India, *Heartbeats and Flow: Exploring Percolation Theory in Cardiac Cells*

2023 Behala College, India, *Study of Heart and Brain-like Cells in the Light of Percolation Theory* (Invited Talk)

2022 ICTS-TIFR, India, *Directed LPP in Traffic Model* (Joint Talk)

Academic Service

2025 Celebrando La Fisica 2.0, Organizing Committee

2024 Infinitesimal to Infinite, Co-organizer
2024 Celebrando La Fisica, Organizing Committee

Honours

2019 CSIR–UGC NET (JRF)
2020 GATE Qualified

Technical Skills

Programming Python, Julia, C, C++, Fortran
Scientific Software MATLAB, Mathematica, LaTeX

Languages

Bengali (native)
English (professional)
Hindi (professional)
Interested in learning German

Personal Interests

- Science communication and public engagement
- Reading novels, history and biographies
- Travel and cultural exchange
- Music: guitar player and currently learning the flute
- Swimming and outdoor activities